

Modeling and Simulation in Life Sciences

Population Models

1 Ecological Models

1.1 Single Species Systems

Remember that $\frac{dN}{dt}$ is change in N per unit time

Exponential Growth If b = birth per person per time and d = deaths per person per time, then let $r = b - d$.

$$\frac{dN}{dt} = rN$$

Solution:

$$N(t) = N_0 e^{rt}$$

What happens to this population?

Logistic Growth

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right)$$

What happens to this population?

Most equations can't be solved easily.

So instead, we solve for equilibria.

$$\begin{aligned}\frac{dN}{dt} &= rN \left(1 - \frac{N}{K}\right) \\ \frac{dN}{dt} &= 0 \\ rN^* \left(1 - \frac{N^*}{K}\right) &= 0 \\ N^* = 0 \quad or \quad N^* &= K\end{aligned}$$

Stability of Equilibria

First evaluate the stability of $N^* = 0$.

Near $N^* = 0$,

$$\frac{dN}{dt} \approx rN$$

So as N increase, $\frac{dN}{dt}$ grows exponentially.

Therefore, $N^* = 0$ is an unstable equilibrium.

Stability of Equilibria What happens near $N^* = K$?

Stability near K If N is just slightly above K ,

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) < 0$$

but if N is just slightly below K ,

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) > 0$$

Therefore, $N^* = K$ is stable.

Other Alternatives

- Gompertz Equation

$$\frac{dN}{dt} = r_0 e^{-\alpha t} N$$

- Delay or lag time

$$\frac{dN}{dt} = F(N(t), N(t-T))$$

- Allee effect

$$\frac{dN}{dt} = rN \left(1 - \frac{N}{K}\right) \left(\frac{N}{K_0} - 1\right)$$

1.2 Interacting Populations

Analyzing Interacting Populations

- Nondimensionalize the system
 - reduce the number of parameters
 - simplify the system
- Solve for equilibria
- Analyze stability of equilibria
- Translate back to determine biological significance

Classic Predator-Prey

Lotka-Volterra Predator-Prey Model

$$\begin{aligned}\frac{dN}{dt} &= rN - cNP \\ \frac{dP}{dt} &= bNP - mP\end{aligned}$$

More Realistic Predator-prey models

$$\begin{aligned}\frac{dN}{dt} &= rN \left(1 - \frac{N}{K}\right) - P \left(\frac{A}{N+B}\right) \\ \frac{dP}{dt} &= eP \left(\frac{A}{N+B}\right) - dP\end{aligned}$$

Another More Realistic Predator-prey models

$$\begin{aligned}\frac{dN}{dt} &= rN \left(1 - \frac{N}{K}\right) - P \left(\frac{AN}{N^2 + B^2}\right) \\ \frac{dP}{dt} &= eP \left(\frac{AN}{N^2 + B^2}\right) - dP\end{aligned}$$

Competition

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1}{K_1} - b_{12} \frac{N_2}{K_1}\right) \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2}{K_2} - b_{21} \frac{N_1}{K_2}\right)\end{aligned}$$

Mutualism

$$\begin{aligned}\frac{dN_1}{dt} &= r_1 N_1 \left(1 - \frac{N_1}{K_1} + b_{12} \frac{N_2}{K_1}\right) \\ \frac{dN_2}{dt} &= r_2 N_2 \left(1 - \frac{N_2}{K_2} + b_{21} \frac{N_1}{K_2}\right)\end{aligned}$$

2 Examples from Epidemiology

Divide population up into distinct classes

- S = Susceptibles
- I = Infectives
- R = Recovered

Classes used depend on disease dynamics

SIR Model - constant population

$$\begin{aligned}\frac{dS}{dt} &= \mu(S + I + R) - \beta SI - \mu S \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I \\ \frac{dR}{dt} &= \gamma I - \mu R \\ N &= S + I + R\end{aligned}$$

$S(0) = S_0, I(0) = I_0, R(0) = 0$. All parameters are assumed to be positive.

Questions for Epidemic Models Given all parameters and initial conditions

- Does the infection spread or die out?
- If it does spread, how does it develop with time?
- When will it start to decline?

Solve for Equilibria

Note: Since N is a constant, we can solve for only S and I , then if we need R , we can calculate it easily.

$$\begin{aligned}\frac{dS}{dt} &= \mu N - \beta SI - \mu S = 0 \\ \frac{dI}{dt} &= \beta SI - \gamma I - \mu I = 0\end{aligned}$$

Gives two equilibria:

$$\begin{aligned}S^* &= N, \quad I^* = 0 \\ S^* &= \frac{\gamma + \mu}{\beta}, \quad I^* = \frac{\mu(\beta N - \gamma - \mu)}{\beta(\gamma + \mu)}\end{aligned}$$

R_0 Basic reproduction rate

- R_0 is defined to be the number of secondary infections produced by one primary infection in a wholly susceptible population.

- So if $R_0 > 1$, then the disease will spread.
- For SIR model, R_0 is calculated by linearizing the equation for $\frac{dI}{dt}$ about $I = 0$, which we have already done.
- So the criteria for determining if the epidemic will spread is,

$$R_0 \equiv \frac{\beta N}{\gamma + \mu}.$$